

1. Administrative & Study Identification

Full Study Title: A Randomized, Participant- and Investigator-blinded, Placebo-controlled, Parallel Group Study to Assess the Efficacy, Safety and Pharmacokinetics of EYU688 in Patients With Dengue Fever **Short Title/Acronym:** EYU688 Efficacy and Safety Study in Dengue Fever **Protocol Number:** CEYU688A12201 **NCT Number:** NCT06006559 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** EYU688 (NITD-688, an oral pan-serotype NS4B inhibitor) **Phase of Development:** Phase 2 (Phase 2a) **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To characterize the effect on dengue viral load, fever clearance time, and clinical signs and symptoms with the treatment of EYU688 compared with placebo in patients with dengue fever. **Secondary Objectives:** To assess the time needed between fever onset and defervescence, evaluate the incidence and severity of adverse events (AEs), and characterize the pharmacokinetic (PK) profile of EYU688 using intensive and sparse sampling cohorts.

2.2 Background & Rationale Dengue fever is a growing global health threat with no specific antiviral drugs currently approved for treatment. Management is predominantly supportive. EYU688 is an investigational, orally administered small molecule that targets the dengue virus non-structural protein 4B (NS4B). By disrupting viral replication early in the disease course, EYU688 has the potential to accelerate viral clearance, reduce the duration of clinical symptoms, and prevent progression to severe disease.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Participant- and Investigator-blinded) **Randomization:** Randomized, Parallel Group

3.2 Planned Enrollment & Duration Target \$N\$: 108 adult patients **Number of Sites:** Multicenter global study (e.g., India, Malaysia) **Estimated Study Start:** 20-Feb-2024 **Estimated Primary Completion:** 07-Jan-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female, 18 - 60 years old (inclusive). 2. History or presence of fever ($\geq 38^{\circ}\text{C}$) with onset of fever ≤ 48 hours prior to treatment start. 3. At least one of the following criteria indicating dengue infection: Nausea or vomiting; presence of rash, aches, or pains including headache, muscle, or joint pain. 4. Positive test for dengue fever.

4.2 Exclusion Criteria 1. Pregnant or nursing (lactating) women. 2. Usage of any anticoagulant drugs or Proton Pump Inhibitors (PPIs) up to 48 hours prior to screening. 3. Clinical signs and symptoms of severe dengue according to the WHO 2009 Dengue Guidelines at screening. 4. Significant laboratory abnormalities (e.g., Platelet count $< 80,000/\text{mm}^3$, Absolute neutrophil count $< 1500/\mu\text{L}$, ALT/AST $> 1.5 \times \text{ULN}$) or ECG abnormalities (QTcF > 450 msec for males or > 460 msec for females).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Oral EYU688 versus matching Placebo. Participants are divided into two parallel cohorts due to different PK sampling schedules (intensive PK [cohort 1] and sparse PK sampling [cohort 2]). **Route of Administration:** Oral (Hard Gelatin Capsule). **Duration of Treatment:** Short-term antiviral administration for acute infection.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for dengue fever (e.g., fluid replacement, protocol-approved antipyretics). **Prohibited:** Anticoagulant medications, PPIs, and any other medications likely to have a significant drug-drug interaction with EYU688.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day 0 (≤ 48 h of fever)	Informed Consent, Dengue Testing, Vital Signs, ECG, Safety Labs
Baseline	Day 1	Randomization, First Dose, Baseline Viremia
Interim	Days 2 - 7	Intensive/Sparse PK blood draws, Efficacy Assessment (48-hour viremia check), Fever Monitoring
End of Study	Follow-up	Final Safety Labs, AE/SAE Assessment, ECG, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy assessment quantifying viremia reduction at 48 hours post-treatment start from baseline.

Measurement Method: Central/Local Laboratory Virological Analysis (DENV RNA viral load).

7.2 Secondary & Exploratory Endpoints * Time needed between fever onset and defervescence (fever clearance time). * Incidence and severity of adverse events (AEs) by treatment group, including changes in vital signs, ECGs, and clinical laboratory results. * Pharmacokinetic (PK) parameters of EYU688.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous inpatient/outpatient monitoring of clinical signs, vital signs, ECGs, and hematology/chemistry parameters. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review committee established by Novartis to oversee trial safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy outcomes; PK Analysis Set for intensive/sparse cohorts; Safety Set for AE evaluations. **Sample Size Justification:** Target enrollment of $N=108$ subjects, statistically powered to detect a significant $\log_{10}$ reduction in DENV viremia at 48 hours compared to placebo. **Handling of Missing Data:** Standard multiple imputation or mixed-effects models for repeated measures (MMRM) for continuous virological endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, accurate PK sampling, and safety reporting. **Audit Trail:** Secure electronic Case Report Forms (eCRF) with stringent data clarification processes and source document verification.

11. Study Identifiers & Governance

Primary ID: CEYU688A12201 **Registry Number:** NCT06006559 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** EYU688 in Patients With Dengue Fever **Official Regulatory Title:** A Randomized, Participant- and Investigator-blinded, Placebo-controlled, Parallel Group Study to Assess the Efficacy, Safety and Pharmacokinetics of EYU688 in Patients With Dengue Fever

12. Clinical Framework (Dengue Specific)

Primary Condition: Dengue Fever **Diagnosis Threshold:** Fever $\geq 38^{\circ}\text{C}$ (onset ≤ 48 hours), positive diagnostic test, and ≥ 1 clinical symptom (e.g., nausea, rash, myalgia). **Medical Methodology:** Direct-acting antiviral (NS4B inhibitor) + Standard Supportive Care. **Optimization Target:** Rapid viremia reduction at 48 hours and accelerated defervescence.

13. Study Procedures & Methodology

Management Pathway: Acute viral infection management and structured PK sampling. **Education Protocol (HCP):** Protocol-specific training for intensive vs. sparse PK sampling cohorts and stringent ECG monitoring. **Education Protocol (Patient):** Acute symptom reporting, antipyretic guidelines, and mandatory contraception adherence. **Quality Audit Mechanism:** Source data verification for virological clearance metrics and stringent AE documentation.

14. Observation & Follow-Up Schedule

Total Duration: Acute treatment phase + post-treatment follow-up (approx. 2-4 weeks). **Onsite Visit Cadence:** Intensive daily monitoring during the acute febrile phase (Days 1-7). **Remote Monitoring Frequency:** As necessary during the convalescent follow-up period. **Checklist Review Interval:** End of active study and final follow-up visit.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				